

Non-Specific Protein Modifications by a Phytochemical Induce Heat Shock Response for Self-Defense

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Accumulated evidence shows that some phytochemicals provide beneficial effects for human health. Recently, a number of mechanistic studies have revealed that direct interactions between phytochemicals and functional proteins play significant roles in exhibiting their bioactivities. However, their binding selectivities to biological molecules are considered to be lower due to their small and simple structures. In this study, we found that zerumbone, a bioactive sesquiterpene, binds to numerous proteins with little selectivity. Similar to heat-denatured proteins, zerumbone-modified proteins were recognized by heat shock protein 90, a constitutive molecular chaperone, leading to heat shock factor 1-dependent heat shock protein induction in hepa1c1c7 mouse hepatoma cells. Furthermore, oral administration of this phytochemical up-regulated heat shock protein expressions in the livers of Sprague-Dawley rats. Interestingly, pretreatment with zerumbone conferred a thermoresistant phenotype to hepa1c1c7 cells as well as to the nematode *Caenorhabditis elegans*. It is also important to note that several phytochemicals with higher hydrophobicity or electrophilicity, including phenethyl isothiocyanate and curcumin, markedly induced heat shock proteins, whereas most of the tested nutrients did not. These results suggest that non-specific protein modifications by xenobiotic phytochemicals cause mild proteostress, thereby inducing heat shock response and leading to potentiation of protein quality control systems. We considered these bioactivities to be xenohormesis, an adaptation mechanism against xenobiotic chemical stresses. Heat shock response by phytochemicals may be a fundamental mechanism underlying their various bioactivities.